

PROJECT DEVELOPMENT REPORT

Food Ordering and Customer Trends Analytics Phase

Date	14 July 2026	Name	Bhoomi Jain
Project Name	Food Ordering And Customer Trends		
Maximum Marks	4 Marks		

Project Development Phase

Data Preprocessing Steps

Step 1: Import the Dataset

- Imported the Food Ordering Behaviour dataset into Tableau Public.
- Verified that all records and fields (Order ID, Customer Age Group, Cuisine, Meal Type, City, Revenue, Customer Rating, Delivery Time, Restaurant Segment, etc.) were imported successfully.
- Confirmed that the dataset was ready for preprocessing.

Step 2: Check for Missing Values

- Examined the dataset for null or blank values.
- Verified that important fields such as Revenue, Customer Rating, Cuisine, Meal Type, and City contained valid information.
- Missing values were handled appropriately to ensure accurate analysis.

Step 3: Remove Duplicate Records

- Checked the dataset for duplicate order records using Order ID.
- Removed duplicate entries to prevent inaccurate counts and incorrect business insights.
- Ensured that every order was represented only once.

Step 4: Validate Data Types

Ensured that each attribute had the correct data type:

```
Order ID → String
Revenue → Number
Customer Rating → Number (Decimal)
Delivery Time → Number
Customer Age Group → Categorical
Cuisine → Categorical
Meal Type → Categorical
Restaurant Segment → Categorical
City → Geographic / String
```

Step 5: Verify Data Consistency

- Checked for invalid or unrealistic values such as negative revenue, invalid customer ratings, or incorrect delivery times.
- Standardized cuisine names, city names, and restaurant segment values.
- Ensured consistency across all categorical fields before visualization.

Step 6: Prepare Calculated Fields

Created calculated fields required for dashboard analysis, including:

- Total Orders
- Total Revenue
- Average Customer Rating
- Average Delivery Time
- Revenue by Cuisine
- Revenue by City
- Order Count by Meal Type (KPI Measures for Dashboard)

Step 7: Final Clean Dataset

- Verified that the cleaned dataset was complete and suitable for analysis.
- Confirmed that all preprocessing activities were completed successfully.
- The dataset was finalized for dashboard creation and business intelligence visualizations in Tableau Public.

Business Questions & Visualization Mapping

S.No.	Business Question	Visualization Technique
1	How many food orders are available for analysis after data preprocessing?	Total Order Count (KPI Card)
2	What is the total revenue generated from food orders?	Total Revenue (KPI Card)
3	Which cuisine category receives the highest number of customer orders?	Cuisine Demand Analysis
4	How does revenue vary across different cities?	City-wise Revenue Analysis
5	What is the average customer rating across restaurant segments?	Customer Rating Analysis
6	How does average delivery time vary among different restaurant segments?	Delivery Time Analysis
7	What are the ordering patterns across different meal types?	Meal Type Analysis
8	How does customer ordering behaviour differ across various age groups?	Customer Age Group Analysis

Dashboard Visualizations & Metrics Summary

Key Performance Indicators (KPI Cards)		
TOTAL ORDERS 50,000	TOTAL REVENUE ₹27,386,555.00	AVG ORDER VALUE ₹547.73

Delivery Time Distribution Analysis

A breakdown of orders based on distinct counts of Order IDs distributed across delivery times ranging from 0 to 14 minutes, illustrating consistency and delivery performance thresholds across segments.

Customer Rating & Meal Type Analysis

Analyzed via average ratings across distinct meal categories: **Breakfast, Snacks, Lunch, and Dinner**, pinpointing customer satisfaction levels across dining periods.

Orders by City & Time Matrix

Comprehensive distribution highlighting key activity periods (Morning, Afternoon, Evening, Night) across major metro areas including **Mumbai, Pune, Bangalore, Delhi, Hyderabad, and Chandigarh**.

Cuisine Demand & Demographic Distribution

A visualization mapping revenue and total volume across primary cuisines (Biryani, Chinese, Desserts, Fast Food, North Indian, South Indian) alongside the distribution of customer age tracking groups (ranging from ages 18 to 44).